Optimized Classification of Coronary Artery Disease Using Feature Engineering and One-Dimensional Convolutional Neural Network

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***Abstract*—Coronary artery disease (CAD), which is one of the leading causes of the global morbidity and mortality rates, needs for early, accurate, and non-invasive diagnostic methods. Electrocardiogram (ECG) signals have been used to obtain information about the electrical activity of the heart and are considered a potentiality in modality for detecting CAD. This paper proposes an enhanced diagnostic methodology that integrates advanced feature extraction and a One-Dimensional Convolutional Neural Network (1D-CNN) for automatic CAD classification. The PTB-XL ECG recordings dataset, which contains annotated multi-lead ECG signals with standardized diagnostic statements, has been used in this study. A binary classification task has been performed to group the ECGs of coronary artery disease related classifications to a CAD class, while all other ECGs classified as normal are assigned to a control class. The dataset featured class imbalance resembling real clinical distribution with 90% Normal and 10% CAD segments. The proposed model achieved an accuracy of 95.09% and an AUC score of 0.958. The results demonstrate that the proposed method can facilitate accurate ECG-based CAD detection, minimize the chance of manual misinterpretation, and enable real-time clinical screening and early diagnosis in a scalable manner.**

***Index Terms*—Coronary Artery Disease, Electrocardiogram (ECG), One-Dimensional Convolutional Neural Network**

I. Introduction

Cardiovascular diseases (CVDs) are a major public health concern, being the leading cause of death globally, and coronary artery disease (CAD) is the most common type, contributing to a total of about 17.9 million deaths per year [1]. CAD is a condition in which the coronary arteries are narrowed by a layer of atherosclerotic material that may significantly reduce blood flow to the myocardium and cause life-threatening events, such as myocardial infarction [2], [3]. Prompt and precise diagnosis of CAD is important to guide timely treatment and improve outcomes. Conventional diagnostic techniques, such as coronary angiography, are invasive and costly making them inapplicable for population screening. Therefore, it is necessary to develop non-invasive, low-cost, and automated diagnostic techniques. Since electrocardiography (ECG) is non-invasive and easily accessible, rich in information regarding the electrical activity of the heart, it has become an attractive technique for detection of CAD [4]. Nevertheless, visual analysis of ECG signals require expertise and is susceptible to human errors and inter-observer variability. These challenges have promoted the research on automated systems for CAD detection based on machine learning and deep learning techniques.

The recent developments in deep learning have transformed the ECG analysis allowing automatic feature extraction from the raw signals [4], [6]. Convolutional neural networks (CNNs), particularly one-dimensional CNNs (1D-CNNs), have been proven to be very effective in capturing temporal and morphological features of ECG waveforms [7], [8].

However, there are still challenges related to signal quality estimation, noise reduction and feature optimization. This work overcomes these limitations by using the advanced feature engineering techniques, such as sample entropy and standard normalization and a developed 1D-CNN model for high accuracy and computational efficiency for CAD classification from ECG signals based on the PTB-XL dataset [11].

II. Literature Review

A notable amount of work has focused on ECG preprocessing and quality improvement as the primary step for

CAD detection. Acharya et al. [4] conducted an effective study to ascertain the influence of ECG segment length on the performance of the CAD detection, and showed that the longer segments embody more rich temporal dependencies with the computational complexity. Their preprocessing stage consists of a bandpass filtering to remove baseline wander and high-frequency noise. Similarly, Phoemsuk and Abolghasemi [7] emphasized that it is necessary to eliminate the non-informative part of the ECG (captured by a group of statistical features including standard deviation and sample entropy) for improving quality of the signal and preventing overfitting through redundant model training. Research like Holgado-Cuadrado et al. [15] went ahead to examine noise specifications of ECG signals, differentiating artifacts originating from muscle movement, electrode movement, baseline wander, powerlines interference among others and emphasized in identifying adaptive filtering techniques. Choi et al. [19] showed that standardized pipelines for preprocessing (using Butterworth bandpass filter and normalization) improve model generalizability among clinical datasets.

Feature extraction is also important in early CAD detection system besides preprocessing. Traditional machine learning methods, such as the enhanced support vector machine (SVM) framework presented by Dolatabadi et al. [5], made use of handcrafted time-domain and frequency-domain features such as statistical moments and spectral descriptors to achieve promising diagnostic accuracy. To further enhance the quality of features, Phoemsuk and Abolghasemi [7] applied an entropy-based quality gating method that ensures that only physiologically meaningful ECG segments to participate in classification. Advanced signal decomposition techniques have been explored by Sharma et al. [13] using wavelet filter banks to obtain discriminative time-frequency features for CAD detection. Alcaraz and Rieta [16] also contributed to the theoretical background of entropy-based feature extraction, and they demonstrated optimal parameter configurations for sample entropy calculation in cardiac signals. In the recent years, hybrid feature engineering methods combining handcrafted statistical features and deep-learned representations have shown the feasibility to exploit multi-view complementary information for more robust classification [17].

The advent of deep learning has shifted CAD detection research toward end-to-end systems that can learn discriminative representations from raw ECG signals. Acharya et al. [4] introduced one-dimensional convolutional neural networks (1D-CNNs) for automated detection of CAD with a demonstration that convolutional layers can effectively learn morphological ECG patterns, including QRS complexes and ST-segment deviations. Phoemsuk and Abolghasemi [7] presented concise CNN models for single-lead ECG classification, whereas Bütün et al. [8] proposed the capsule-based CNN to preserve spatial hierarchies in ECG waveforms. Multi-scale and multi-granularity CNN models have been proposed to model short-term and long-term temporal dependencies as well [18]. Moreover, biologically motivated convolutional kernels, such as Gabor filters, have been used to increase feature selectivity and enhance interpretability in ECG analysis [22]. Hybrid learning techniques for CNNs have shown additional increase in performance by capturing architectural diversity [24].

For more explicit modeling of temporal dependencies, hybrid deep learning models based on CNNs and RNNs have been extensively studied. It has been demonstrated that the CNN-LSTM and CNN-BiLSTM models can efficiently learn local morphological information and sequential heartbeat dynamics, resulting in superior classification accuracy [6], [9], [10].

Performance evaluation and clinical verification always play an important role in CAD classification studies. The previous works have used a number of evaluation metrics such as accuracy, sensitivity, specificity, F1-score, confusion matrix, and area under the ROC curve for determining the diagnostic effectiveness [4], [5], [19]. Stratified cross-validation and inter-patient validation protocols have been stressed to minimize subject-specific bias and to be representative [5]. Clinical studies, validated against coronary angiography results, have illustrated the practical tradeoff between the sensitivity of early detection and false-positive rates in real-life screening [19]. The results show that ensemble-based diagnosis systems provide statistically significant improvements over single-model baselines, emphasizing the need for strong evaluation protocols [23], [25].

Nevertheless, there still exist some restrictions in the above approaches. Most of the previous methods rely on complicated multi-stage frameworks or large handcrafted feature pipelines, which are unscalable and difficult to be applied in real-time. Moreover, there is a lack of research on entropy-based ECG segment selection in conjunction with lightweight one-dimensional convolutional neural network (1D-CNN) models for accurate and reliable CAD screening. These observations have provided the motivation for the current work, which employs statistically inspired feature engineering with a constrained 1D-CNN architecture for high precision, scalable and clinically practical CAD detection from ECG signals.

III. Methodology

This section describes the framework for automated coronary artery disease (CAD) detection using electrocardiogram (ECG) signals. The methodology consists of data acquisition, signal preprocessing and segmentation, feature engineering, class label assignment, deep learning-based model design, training, and performance evaluation. Fig. **??** shows the entire workflow of the proposed CAD classification system. Each stage is designed to enhance the reliability of the signal, extract discriminative temporal features, and enable accurate and scalable CAD detection using a lightweight one-dimensional convolutional neural network (1D-CNN).

A. Data Acquisition

The proposed model uses the ECG signals from PTB-XL dataset, a large-scale, publicly available clinical dataset [11]. The dataset consists of 12-lead ECG recordings of 10 s

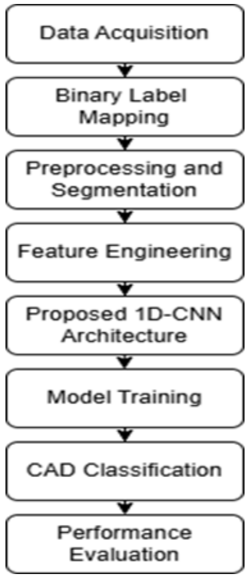

Fig. 1. Overview of the proposed model.

duration, sampled at 500 Hz or 100 Hz and annotated using standardized Statement of Clinical Purpose (SCP) diagnostic codes. ECG records sampled at 500 Hz were selected and all twelve leads were taken into consideration when extracting the signals. As PTB-XL does not contain binary coronary artery disease (CAD) labels, the diagnostic SCP annotations were retained for subsequent binary class formulation. The PTB-XL dataset is widely used in ECG-based CAD detection studies due to its clinical relevance and standardized labeling [4], [6], [7].

B. Binary Label Mapping

The PTB-XL dataset does not include a combined binary label for the existence of coronary artery disease (CAD). Each ECG record is labeled with standardized SCP diagnostic codes indicating clinically relevant ECG abnormalities including myocardial infarction, ischemic changes, arrhythmias or normal sinus rhythm [11].

The ECG records associated with SCP codes related to ischemia and myocardial infarction are grouped into a single CAD class, whereas ECGs labeled as normal are assigned to the Normal class. In particular, SCP diagnostic statements such as anterior septal myocardial infarction (ASMI), inferior myocardial infarction (IMI), ischemic changes (ISC), and ST–T abnormalities (STTC) are indicated as CAD. ECGs with the NORM SCP code are considered to be healthy control samples. This type of categorization has also been applied in previous binary ECG-based CAD detection studies to retain binary classification while maintaining clinical relevance [4], [7], [11].

These binary labels are saved into a meta-file and used throughout the pipeline. Each ECG record is assigned a ground-truth label of CAD or Normal according to the SCP code groupings. In the preprocessing stage, the entire ECG segment is extracted from a record. Consequently, all segments extracted from a given record possess the corresponding record-level label, which facilitates learning at the segment level while retaining diagnostic information at the subject level. This SCP-based labeling approach addresses the missing explicit CAD labels in PTB-XL while providing a meaningful and reproducible CAD vs. Normal classification task.

C. Preprocessing and Segmentation

The raw ECG signals recorded from the PTB-XL dataset undergo a preprocessing pipeline for quality enhancement and noise artifact removal. For each ECG record, a fourth order Butterworth band-pass filter with cut-off frequencies of 0.5–40 Hz is used to reject the baseline wander and high-frequency noise while keeping the clinically meaningful cardiac contents (P-wave, QRS-complex, and ST–T-segment). Band-pass filtering in this particular frequency range is a common practice in ECG-based CAD analysis and has been reported to improve the result of the subsequent classification [4], [11], [15].

The PTB-XL recordings are provided at various sampling rates; therefore, all ECG signals are resampled to a common sampling rate of 250 Hz for uniformity in the dataset. Since each ECG recording has twelve leads, to reduce the inter-lead redundancy and computational complexity, all lead signals are averaged to yield a single representative lead per recording. The same lead averaging strategy was used in previous ECG classification works to preserve global morphology and allow for lightweight model design [4], [7].

Z-score normalization is applied to the resulting ECG signal to achieve zero mean and unit variance, which makes the learning more stable and also prevents high-amplitude segments from being dominated in the training phase.

$$x_{\text{norm}}(n) = \frac{x(n) - \mu}{\sigma} \tag{1}$$

where $x(n)$ denotes the ECG signal sample, μ is the mean value of the signal, and σ represents the standard deviation. This normalization ensures scale invariance and improves training stability of the deep learning model. The normalized ECG signal is split into fixed-length non-overlapping segments of 1 second duration (i.e., 250 samples per segment). Applying segmentation to long-term ECG recordings not only increases the number of training samples but also enables the deep learning model to capture localized temporal patterns related to coronary artery disease [4], [6].

D. Feature Engineering

Feature engineering is used to enhance the quality of the ECG segments on which the model is trained by removing portions of the signal that are non-informative, noisy, or corrupted. Although the proposed framework is mainly based on deep learning for automatic feature learning, a few simple and effective statistical features are introduced as a quality control before classification. The proposed model allows physiologically meaningful ECG segments to contribute to the learning process, thereby enhancing model robustness and generalization.

Sample Entropy (SampEn) is calculated to measure the complexity and irregularity of ECG segments besides amplitude-based processing. Sample entropy is a popular nonlinear measure of signal predictability in physiological time series and it has been proven to effectively distinguish structured cardiac rhythms from noise contaminated segments [7], [16]. Given a time series $x = \{x_1, x_2, \ldots, x_N\}$, sample entropy is defined as:

$$\text{SampEn}(m, r) = -\ln\left(\frac{A}{B}\right) \tag{2}$$

where m denotes the embedding dimension, r represents the tolerance threshold, A is the number of pairs of template vectors of length $m+1$ that match within tolerance r, and B is the number of pairs of template vectors of length m that match within tolerance r. The embedding dimension is set to $m = 2$, and the tolerance r is defined as a fraction of the standard deviation of the ECG segment, following established recommendations for cardiac signal analysis [16].

Signal quality is evaluated by calculating the standard deviation and sample entropy for each one-second ECG window. Standard deviation quantifies the amplitude variation and detects flat-line or very noisy artifacts, whereas sample entropy quantifies signal complexity. Extreme low variance, extreme high variance segments and entropy values outside the acceptable ranges are rejected, as they are non-informative or dominated by noise. This combined filtering strategy ensures that only physiologically meaningful ECG segments are preserved for the following deep learning-based classification. The filtered ECG segments are then directly input to the proposed 1D-CNN model to learn features and classify automatically. This lightweight feature engineering strategy reduces the effect of artifacts, improves training stability, and lowers computational overhead without relying on extensive handcrafted feature sets.

E. Proposed 1D-CNN Architecture

The filtered one-second ECG segments are classified using a lightweight one-dimensional convolutional neural network (1D-CNN) designed to achieve a balance between the classification accuracy and computational efficiency. The network operates directly on normalized ECG segments of length 250 samples and is optimized for binary classification between Normal and CAD classes.

The proposed model consists of three convolutional blocks arranged hierarchically. Each block consists of a one-dimensional convolutional layer, batch normalization layer and a max-pooling layer. Rectified Linear Unit (ReLU) activation functions are used to introduce non-linearity and to allow effective learning of ECG waveform characteristics, such as QRS morphology and ST–T segment variations. The convolutional layers apply progressively deeper filters to extract temporal features ranging from low-level waveform patterns to higher-level discriminative representations.

Batch normalization is performed at the end of each convolution layer to normalize training and faster convergence, meanwhile, the max-pooling layers down-sample the temporal dimension and computational load. Dropout regularization is added after convolutional and fully connected layers to prevent overfitting, especially considering the large number of ECG segments obtained by signal segmentation.

The learned convolutional feature maps are flattened and passed to a fully connected dense layer, with ReLU activation function. The final output layer is a single neuron that uses sigmoid function and returns a probability score on the presence of coronary artery disease. This design allows for efficient end-to-end learning from ECG segments and at the same time yields a compact model that can be used for scalable and real-time CAD screening. Other comparable lightweight 1D-CNN architectures were also shown to obtain good results in ECG-based CAD classification [4], [7].

F. Model Training

The discriminatory performance of the proposed 1D-CNN model is trained through a supervised learning paradigm where the binary cross-entropy is considered as the loss function and the Adam optimizer is used. The dataset is split into training and test sets by stratified 80–20 split with 80% of the ECG segments used for training and 20% held out for testing. During training, early stopping and model checkpointing are performed so that overfitting can be avoided and the best fit model can be saved based on the validation performance. The above described training process results in stable convergence, and the end-to-end network generalizes well to unseen ECG segments.

G. CAD Classification

The trained 1D-CNN performs the binary classification between Normal and CAD, for which each one-second ECG segment is classified independently at the segment level. For each segment, the sigmoid output neuron provides a probability score representing the probability of having coronary artery disease. Utilizing this segment-level classification enables the model to focus on identifying localized pathological patterns in ECG recordings and facilitates scalable CAD screening without the need for record-level aggregation.

H. Performance Evaluation

The proposed feature-based automated CAD classification technique is assessed based on conventional classification performance metrics, i.e., accuracy, precision, recall, and F1-score. These metrics provide a more comprehensive assessment of the prediction performance of the model, particularly when class imbalance is present. Further, receiver operating characteristic (ROC) analysis and the area under the ROC curve (AUC) are employed to assess the discriminatory power of the model at various decision thresholds.

IV. Results and Discussion

The performance of the proposed CAD detection framework is described and its performance is compared with the classical machine learning models. The evaluation is performed on the PTB-XL dataset with a stratified 80–20 train–test split at the segment level. Accuracy, sensitivity and specificity of our proposed model are evaluated and the receiver operating characteristic area under the curve (ROC-AUC) is reported in order to have an enhanced evaluation in the presence of class imbalance.

A. Preprocessing Validation

To verify the preprocessing procedure, the ECG signals before and after band-pass filtering and normalization are presented in Fig. **??**. The baseline wander and high frequency noise are reduced in the filtered signal, and also keeps the clinically relevant waveform components (such as P-wave, QRS complex and ST segment). This demonstrates the enhancement of the signal quality in the preprocessing prior to segmentation and model training.

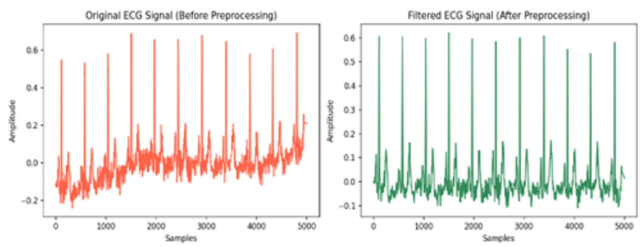

Fig. 2. ECG signals before and after preprocessing.

B. Performance Comparison

As shown in Table I, the proposed 1D-CNN achieves the highest classification accuracy (95.09%) among all evaluated models. The model also demonstrates improved sensitivity (0.661) compared to classical machine learning approaches, indicating better detection of CAD-positive cases. Whilst all models display high specificity, the proposed method achieves a more harmonic result. The superior discriminative performance predicted with the proposed method is also consistent with the ROC-AUC values.

TABLE I
PERFORMANCE COMPARISON OF CAD DETECTION MODELS

Model	Accuracy (%)	Sensitivity	Specificity	ROC-AUC
Logistic Regression	90.16	0.261	0.987	0.816
SVM (RBF Kernel)	91.46	0.340	0.991	0.825
Random Forest	92.09	0.463	0.982	0.891
Proposed 1D-CNN	**95.09**	**0.661**	**0.989**	**0.958**

C. ROC Analysis

The discriminative performance of the evaluated models is further analyzed using ROC curve analysis. Fig. 3 reveals the ROC curves of all evaluated models. The proposed 1D-CNN obtains the best ROC-AUC of 0.958, which is much better than traditional methods, like logistic regression, SVM and Random Forest. This result emphasizes the power of end-to-end deep feature learning from raw ECG segments for effective CAD classification in the presence of class imbalance.

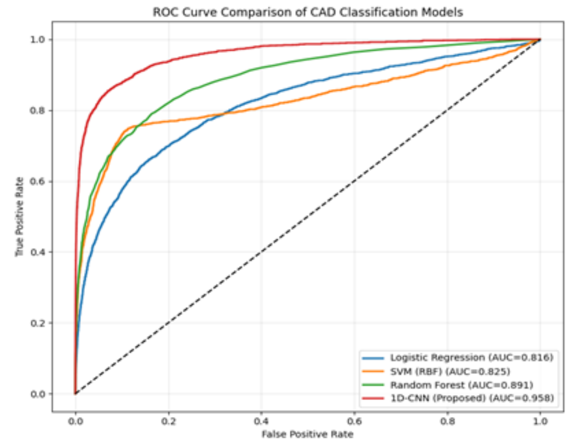

Fig. 3. ROC curve comparison of different CAD classification models.

D. Confusion Matrix Analysis

The confusion matrix analysis also confirms the accuracy of the proposed model. The very high specificity of 98.94% and the overall accuracy of 95.09% indicate a good performance in recognizing the normal ECG segments, but the sensitivity of 66.06% shows the difficulty of identifying CAD by using ECG signals only. Because of the unbalanced dataset, the model is oriented to the specificity to minimize the false positive. Nevertheless, the high ROC-AUC demonstrates that sensitivity can be further improved through threshold adjustment, allowing the model to be efficiently utilized for clinical screening purpose. The increasingly superior performance of the proposed 1D-CNN compared to classical ML models illustrates the benefit of deep temporal feature learning over handcrafted statistical features for detecting subtle ECG changes related to CAD.

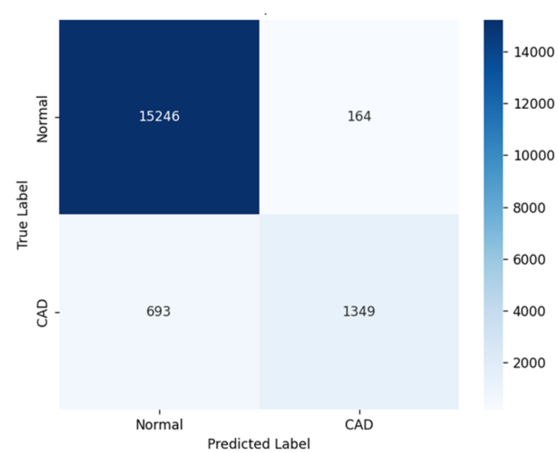

Fig. 4. Confusion matrix of proposed 1D-CNN model.

V. Conclusion

The proposed model presented an automated framework for coronary artery disease (CAD) detection using ECG signals based on a lightweight one-dimensional convolutional neural network. The proposed methodology combines signal preprocessing, statistical feature-based segment filtering, clinically motivated class label assignment, and end-to-end deep learning for accurate CAD classification. The experimental results on the PTB-XL dataset show that the proposed 1D-CNN model has better performance than classical machine learning models in terms of accuracy, ROC-AUC, sensitivity with a relatively high specificity. The results suggest that learning temporal features deeply from the segments of ECG enables the robust and scalable detection of CAD under class imbalance.

In the future, we plan to extend the proposed framework by employing ECG-level decision fusion methods to integrate segment-level predictions for patient-level diagnosis. Further improvements could include addressing data imbalance via cost-sensitive learning or adaptive threshold selection to enhance sensitivity in clinical screening applications. In addition, the use of multi-lead representations and investigation of hybrid deep learning architectures could possibly lead to further enhanced diagnostic performance and generalization on different ECG databases.

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